Feasibility Assessment of Balanced-Model Electric Vehicle Charging-Flexibility Schedules in Unbalanced Three-Phase Distribution Networks

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***Abstract*—Electric-vehicle (EV) charging-flexibility schedules generated with balanced or simplified distribution-network models may violate voltage, phase-unbalance, or equipment limits when implemented in an unbalanced three-phase feeder. This paper evaluates the network feasibility of precomputed vehicle-level unidirectional charging-flexibility schedules. Starting from a vehicle-feasible baseline, the charging-power adjustment of a balanced-model target schedule is proportionally scaled while preserving the original vehicle–time power allocation. Each scaling level is evaluated by nonlinear unbalanced three-phase power flow with bus-voltage, voltage-unbalance-factor (VUF), transformer-loading, and feeder-head-loading constraints. A coarse search with 0.02 increments and refinement to at most 0.002 identifies the largest verified feasible scaling level and the first limiting network constraint. Across 46 fifteen-minute unidirectional charging cases on the IEEE 13-node and European 906-bus low-voltage feeders, 32 target schedules cannot be implemented in full. The mean feasible scaling factors are 0.834 and 0.540 for the two feeders, respectively. Among the 32 limited cases, voltage magnitude or VUF is the first limiting constraint in 27 cases, and all 21 limited cases on the 906-bus feeder are voltage-limited. The results show that three-phase network validation can quantify the executable portion of balanced-model EV charging schedules and identify the network constraints that limit their implementation.**

***Index Terms*—electric vehicles, charging flexibility, unbalanced distribution networks, three-phase power flow, feasibility assessment**

I. Introduction

Aggregated electric-vehicle (EV) charging can provide distribution-level flexibility by shifting charging power and charging time. The usable flexibility is determined by both vehicle constraints, such as availability, charging limits, and energy requirements, and network constraints, including bus voltage, phase unbalance, and equipment capacity. Unbalanced distribution-network constraints have therefore been incorporated into aggregate flexibility characterization [1], coordinated EV charging and discharging [2], and dynamic operating-envelope or aggregator operating-region construction [3]–[6]. Data-driven approaches have also been developed to reduce the computational burden of aggregate flexibility characterization [7].

Three-phase network constraints can also be included directly during scheduling or flexibility-boundary calculation. Churkin *et al.* quantify the effects of voltage unbalance and resource coordination on aggregated P–Q flexibility using nonlinear three-phase optimal power flow (OPF) [8]. Grid-aware EV frameworks incorporate distribution-network limits into day-ahead scheduling and real-time charging control [9], while coordinated formulations determine network-secure energy flexibility for EV aggregators [10]. In practical operation, however, an EV charging-flexibility schedule may already have been produced using a balanced or simplified feeder model before it is dispatched to an unbalanced network. The network feasibility of that precomputed vehicle-level schedule then needs to be evaluated in the detailed three-phase feeder.

This paper evaluates balanced-model unidirectional (V1G) charging-flexibility schedules in unbalanced three-phase distribution networks. A vehicle-feasible baseline and a balanced-model target schedule are first established. The vehicle-level power adjustment between the two schedules is then scaled by a common factor α. As α increases from 0 to 1, the magnitude of the scheduled adjustment increases while the vehicle–time power allocation remains unchanged. Every candidate schedule is evaluated by nonlinear OpenDSS three-phase power flow under voltage, voltage-unbalance-factor (VUF), transformer-loading, and feeder-head-loading limits. The assessment therefore determines how far a precomputed charging schedule can be implemented before a network constraint is reached.

The main contributions are: 1) a network-feasibility assessment for precomputed vehicle-level EV charging-flexibility schedules that retains the original vehicle–time power allocation during evaluation; 2) a proportional-scaling and boundary-refinement procedure that quantifies the feasible implementation level and identifies the first limiting network constraint; and 3) a 46-case evaluation on the IEEE 13-node and European 906-bus low-voltage feeders, revealing distinct feasible scaling levels and limiting-constraint patterns across the two test

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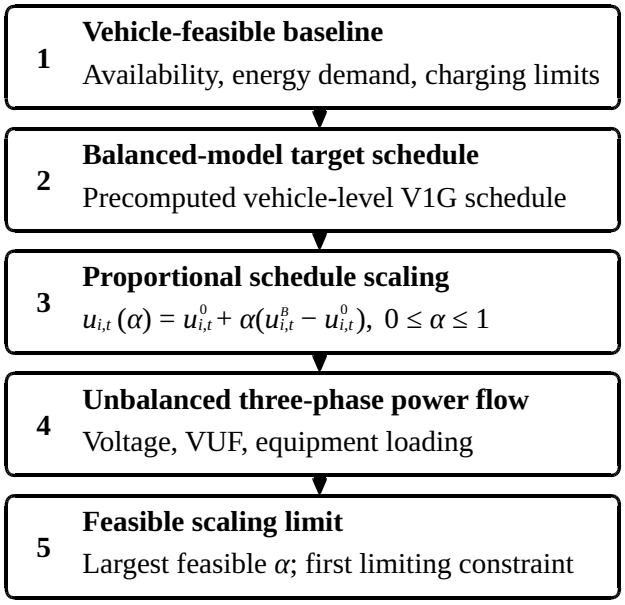


Fig. 1. Network-feasibility assessment procedure.

systems.

II. Network Feasibility Assessment

Consider an aggregation of N EVs over $T = 96$ fifteen-minute time slots. For EV i, the arrival time a_i, departure time d_i, initial energy E_i^0, target energy E_i^{tar}, charging efficiency η_i^c, and charging-power limit are known. Let $u_{i,t}$ denote grid-side charging power in time slot t, with $\Delta t = 0.25$ h. A baseline charging profile uniformly distributes the energy required to reach the target energy over the vehicle availability window:

$$u^0_{i,t} = \frac{E_i^{\mathrm{tar}} - E_i^0}{\eta_i^c \Delta t (d_i - a_i)}, \quad a_i \le t < d_i, \tag{1}$$

with $u^0_{i,t} = 0$ outside the availability window. The baseline is checked against the vehicle availability, charging-power, and departure-energy requirements.

The balanced planning model computes the feasible V1G charging-power increase for a specified 15-min flexibility call. The resulting aggregate target is allocated into a full 96-slot vehicle-level charging plan $u^B_{i,t}$ while satisfying vehicle availability, charging-power, battery-energy, and departure-target constraints. The subsequent network assessment therefore evaluates a complete vehicle-level schedule rather than only an aggregate call-slot set point. Fig. 1 summarizes the assessment procedure.

A. Proportional Schedule Scaling

To evaluate the implementation level of the balanced-model target schedule, the vehicle-level charging adjustment relative to the baseline is scaled by a common factor α:

$$u_{i,t}(\alpha) = u^0_{i,t} + \alpha \left(u^B_{i,t} - u^0_{i,t}\right), \qquad 0 \le \alpha \le 1. \tag{2}$$

At $\alpha = 0$, the candidate schedule equals the vehicle-feasible baseline; at $\alpha = 1$, it equals the complete balanced-model target schedule. Intermediate values proportionally scale every vehicle–time power adjustment relative to the baseline. Thus, increasing α increases the implementation level of the target schedule without selecting different vehicles, reallocating charging times, or recomputing vehicle powers according to three-phase network conditions.

B. Network Feasibility Evaluation

For each candidate α, the corresponding vehicle charging powers are connected to their original buses and phases in the detailed distribution-network model, and nonlinear three-phase power flow is solved in OpenDSS [11]. A candidate schedule is feasible when all energized bus voltages satisfy $0.95 \le V \le 1.06$ p.u., VUF at three-phase buses does not exceed 2%, transformer loading does not exceed 1.0 p.u., and feeder-head loading does not exceed 1.0 p.u. VUF is computed as

$$\mathrm{VUF} = 100\frac{|V_2|}{|V_1|}, \tag{3}$$

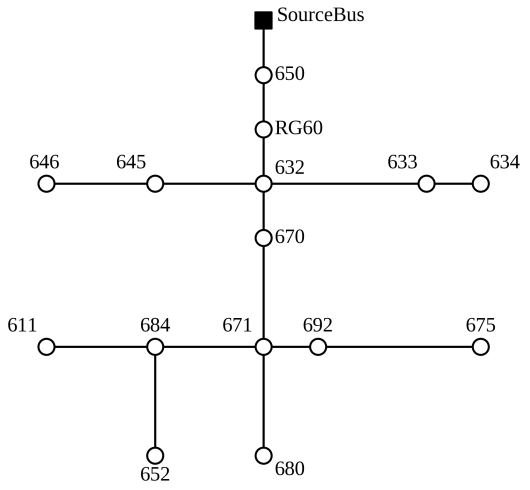
where V_1 and V_2 are the positive- and negative-sequence voltage magnitudes. Transformer loading is the maximum terminal apparent power normalized by the transformer rating. Feeder-head loading is calculated from three-phase active and reactive power using rating bases of 5 MVA for IEEE13 and 0.8 MVA for LV906.

For each case, α is first increased from 0 to 1 in increments of 0.02. If the complete target schedule at $\alpha = 1$ satisfies all network constraints, it is fully feasible. When the first violation is found, the interval between the last feasible point and the first infeasible point is refined until the spacing between adjacent tested values is no larger than 0.002. The last point satisfying all network constraints is recorded as the largest verified feasible scaling factor. For cases with a feasible factor below 1, the immediately following infeasible point is examined to identify the first violated constraint as low voltage, high voltage, VUF, or transformer loading.

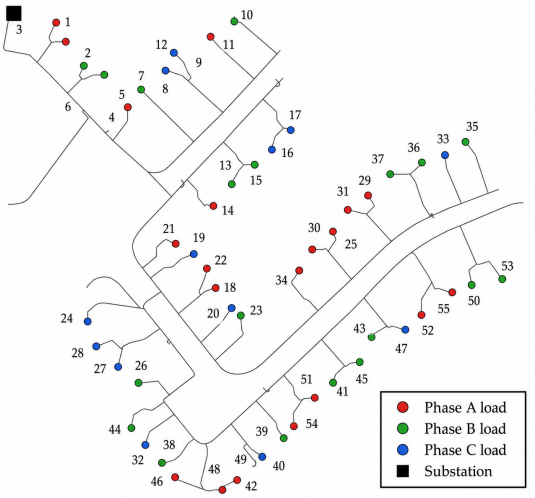
III. Test Systems and Study Cases

The IEEE 13-node test feeder (IEEE13) and IEEE European Low Voltage Test Feeder (LV906) are used for evaluation [12], [13]. Their detailed unbalanced three-phase models are implemented in OpenDSS. For IEEE13, the balanced planning model uses the corresponding positive-sequence representation. For LV906, native phase-specific bus loads are converted to balanced three-phase wye loads while preserving each bus's aggregate active and reactive demand, and EV injections are balanced across phases in the planning model. The network-feasibility evaluation uses the original unbalanced models and restores each EV to its assigned bus and phase. The two test feeders are shown in Fig. 2.

EV inputs include arrival and departure times, initial and target energy, charging-power limits, charging efficiency, and bus/phase assignments. Travel windows are derived from the public 2022 National Household Travel Survey [14]. The study



(a) IEEE 13-node feeder



(b) European 906-bus LV feeder

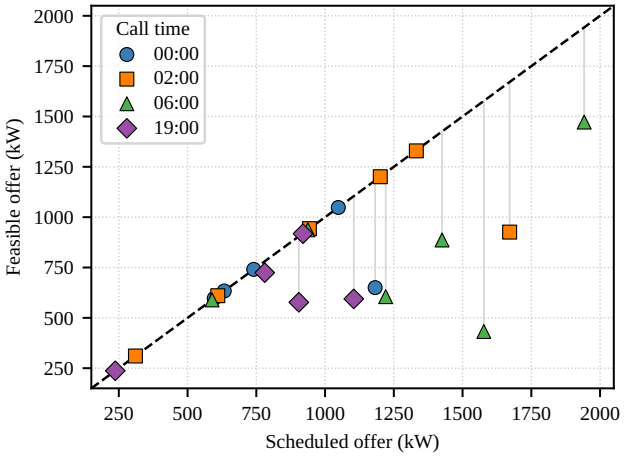
Fig. 2. Test distribution feeders.

TABLE I
V1G STUDY CASES

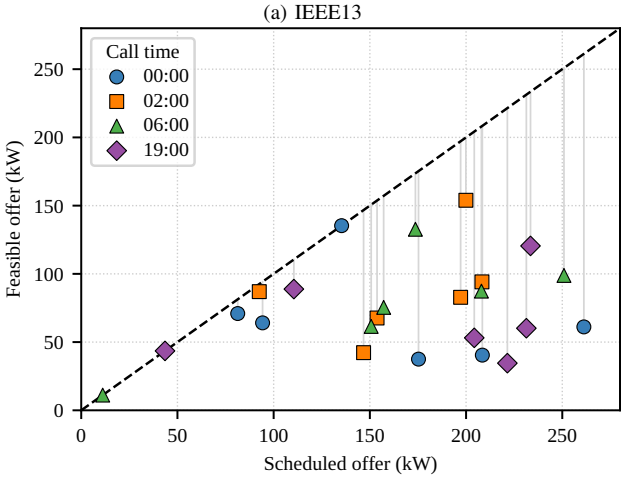
Feeder	Cases	Planning model	EVs/case[a]	Offer (kW)[a]
IEEE13	22	Positive-sequence	173 [65–337]	940 [237–1942]
LV906	24	Bus-balanced 3ϕ	29.5 [11–54]	174 [11–261]

[a]Median [minimum–maximum] across study cases.

considers 15-min V1G flexibility calls that increase charging power at 00:00, 02:00, 06:00, and 19:00. Combining the four call times with six EV scenarios gives 48 candidate feeder–time–scenario combinations. One IEEE13 case is excluded because its baseline violates the detailed network constraints, and one is excluded because a valid balanced-model target schedule is unavailable. The resulting dataset contains 46 cases: 22 on IEEE13 and 24 on LV906. Table I summarizes the case scale.



(a) IEEE13



(b) LV906

Fig. 3. Scheduled and feasible EV charging-flexibility offers.

IV. Results and Discussion

A. Schedule Feasibility

Across the 46 V1G cases, 32 balanced-model target schedules reach a network-feasibility limit before full implementation. Thirty cases have a largest verified feasible scaling factor below 0.95 (65.2% of all cases), and 16 fall below 0.5. The overall mean feasible scaling factor is 0.681 and the median is 0.719.

Fig. 3 translates the scaling result into the call-slot flexibility seen by an operator. For a balanced-model scheduled offer ΔP_{sched}, the corresponding feasible offer is $\Delta P_{\text{feas}} = \alpha \Delta P_{\text{sched}}$. Points on the $y = x$ line represent schedules that can be implemented in full, whereas the vertical distance below the line directly shows the reduction required by the detailed three-phase network. The IEEE13 cases lie predominantly near the full-feasibility line, while many LV906 cases retain only a fraction of the balanced-model offer. Cases with similar scheduled offers can still have substantially different feasible offers, showing that aggregate call magnitude alone does not determine schedule feasibility.

Table II and Fig. 4 summarize the distribution of the feasible scaling factor α. A value of $\alpha = 1$ means that the complete

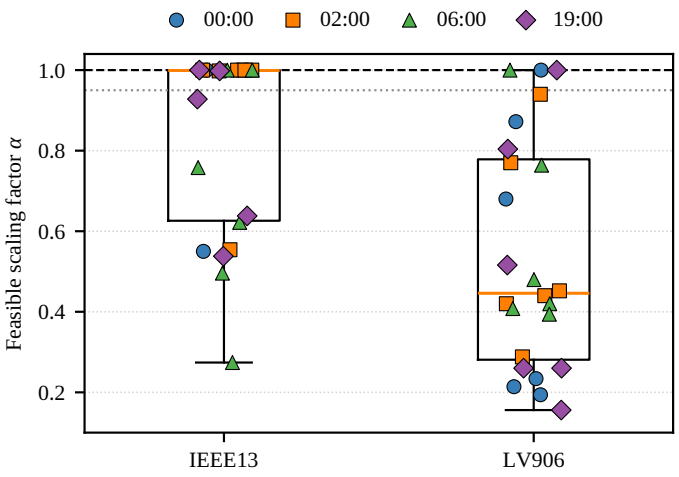


Fig. 4. Feasible scaling factors for the study cases.

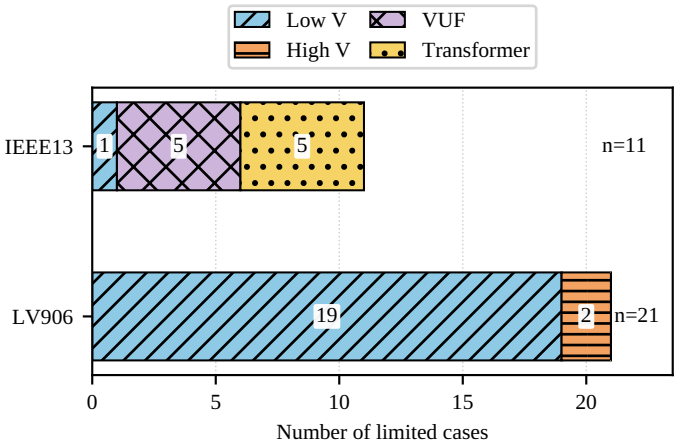


Fig. 5. First limiting network constraints.

balanced-model target schedule satisfies all detailed-network constraints; $\alpha < 1$ means that the scheduled adjustment relative to the baseline must be reduced. The dashed line at 0.95 in Fig. 4 is used as a near-full-implementation reference.

The two feeders show distinctly different feasible scaling levels in the tested cases. IEEE13 has a mean α of 0.834 and a median of 0.999; 11 of its 22 cases implement the complete target schedule. LV906 has a mean of 0.540 and a median of 0.446; only 3 of 24 cases reach $\alpha = 1$. The difference in mean α is 0.294, and a case-level bootstrap gives a 95% interval of [0.144, 0.436] for the IEEE13-minus-LV906 mean difference.

B. Limiting Network Constraints

For the 32 cases with $\alpha < 1$, the first infeasible point immediately after the feasible scaling limit is examined. In all 32 cases, the first network violation occurs in the corresponding 15-min flexibility call slot, directly linking the identified limit to the charging-power increase scheduled for that call.

Fig. 5 classifies the first violated network constraint. Twenty cases are first limited by low voltage, two by high voltage, five by VUF, and five by transformer loading. Voltage magnitude and VUF together account for 27 of the 32 limited cases (84.4%), making voltage-related constraints the dominant first limitation in this study.

The limiting patterns differ between the feeders. Among the 11 limited IEEE13 cases, five are first limited by transformer loading, five by VUF, and one by low voltage. All 21 limited LV906 cases are first limited by bus voltage: 19 by low voltage and two by high voltage. Thus, local voltage feasibility is the dominant limitation for the tested LV906 schedules, whereas the IEEE13 cases exhibit both phase-unbalance and transformer-loading limitations.

TABLE II
LARGEST VERIFIED FEASIBLE SCALING FACTORS

Feeder	n	Mean	Median	$\alpha < 0.95$	$\alpha < 1$
IEEE13	22	0.834	0.999	9	11
LV906	24	0.540	0.446	21	21
Overall	46	0.681	0.719	30	32

The results show that the aggregate charging-flexibility level obtained from a balanced model does not by itself determine the feasibility of the corresponding vehicle-level schedule in a detailed unbalanced feeder. Spatial EV allocation, phase assignment, and the network operating point affect how much of the target schedule can be implemented. Three-phase network validation therefore provides schedule-specific information that is not available from the aggregate balanced-model offer alone.

The numerical study covers two benchmark feeders, 15-min V1G increasing-direction calls, and the voltage, VUF, transformer, and feeder-head loading limits stated above. The two feeders use different balanced planning representations and EV population scales, so the feeder comparison reflects the complete tested case ensembles. Line ampacity is not included in the present network constraints.

V. CONCLUSION

This paper evaluates the network feasibility of vehicle-level V1G charging-flexibility schedules generated by balanced feeder models. Starting from a vehicle-feasible baseline, the target charging adjustment is proportionally scaled and each scaling level is checked by nonlinear unbalanced three-phase power flow. The procedure determines the largest verified feasible scaling factor and the first network constraint that limits further implementation of the precomputed schedule.

Across 46 cases on IEEE13 and LV906, 32 target schedules cannot be implemented in full, with mean feasible scaling factors of 0.834 and 0.540 for the two feeders. Voltage magnitude or VUF is the first limiting constraint in 27 of the 32 limited cases, and all 21 limited cases on the 906-bus feeder are voltage-limited. These results demonstrate the value of validating balanced-model EV charging-flexibility schedules in the detailed unbalanced three-phase network before implementation.

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